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A Dynamic Web Content Reader Application for Low Vision Patients using Python

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Abstract— With six meters highest visibility and 20
degrees

maximum wide view, people who suffer from low vision
are

unable to see words and letters in ordinary newsprint.

This

fact makes the reading process becomes difficult that

can

disturb the learning process and slow the patient's

intellectual

development. Therefore, an application is needed to

help

them read easier. The challenges faced by visually

impaired

people in reading printed text in their day-to-day life are

often

not well understood. This project is based on a

prototype

which helps the user to listen to the contents of the live

websites in Tamil, English, and Hindi etc. It involves the

extraction of content from the websites and translates
audio

output in the languages mentioned above. This is done
using

python libraries with the concept of Natural Language
Processing, Web Mining, Text-to-Speech. It is portable
and

easy to use thus providing a better reading experience
to the

visually challenged people as enjoyed by their sighted
peers.

Key words: Text-to-speech; Python Libraries; Natural
Language Processing; Web Mining; Visually Impaired

Fig. 1: Block Diagram of Text-To-Speech Device

I. INTRODUCTION

Text-to speech device consisted of three main

The complexity of the existing Braille system for the visually

modules, the image processing module, word correction

impaired people is that it requires the text to be translated to

module, and voice processing modules. Image processing

Braille literature. Translating a book or a document into module sets the object position, focus and illumination

Braille literature is a complex, time consuming and more

camera, taking pictures, and converting the image into text.

expensive process. Day-to-day information cannot be

Word correction module makes corrections to the output

translated into Braille literature. To ease this process of image processing module to improve accuracy by matching

reading for the visually impaired people, this prototype has

with Indonesian dictionary. Voice processing module been proposed. Using this prototype, text information can be

changes the writing into sound and process it with specific

converted into its equivalent audio format. This is done using

physical characteristics so that the sound can be understood.

python libraries with the concept of Natural Language

The input image captured by the camera has a size

Processing, Web Mining, Text-to-Speech. It is portable and

of 5 MPI (2592 x 1944 pixels) or 215 ppi(pixels per inch).

easy to use thus providing a better reading experience to the

Based on the specifications of the Tesseract OCR engine, the

visually challenged people as enjoyed by their sighted peers.

minimum character size that can be read is 20 pixels

Remaining section describe about Existing System

uppercase letters. Tesseract OCR accuracy will

decrease with

and Proposed System.

the font size of 10pt.

The image is taken by the user via GPIO pin that are

II. EXISTING SYSTEM connected to the tactile key using interrupt function.

The Existing system done by integrating camera module and Furthermore, the picture is taken by using raspistill program

speakers to Raspberry Pi 3 model B which is a credit card with sharpness mode to sharpen the image. The resulting

sized single board computer. Also, two other software called image has a .jpg format with a resolution of 2592 x 1944

Tesseract and TTS engine (Text to Speech synthesizer) is pixels.

installed to the Raspbian OS. Principally, the camera captures Word correction module gets input from the image

the image and stores it as an image file with a .jpg extension. processing module in the form of text from the image

The OCR engine converts it from image file to text file by processing module. Image processing module can't define

extracting the numbers and characters of the language.

The truth or falsity of the word output, so that the correction

file is given as an input to a python program which gives a module of this word, correction for whole words output from

translated speech output using Google text to speech engine. the image processing requires module. In order to improve

the accuracy of the output image processing module, to

design the word correction module.

The main disadvantage of the existing system is highly expensive. And the portability of the Device is very

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difficult. The Existing system using only converts the printed

content. Also, the capturing paper size is A4 only. The Device

Takes more time to convert the scanned Documents. If any

problem occurred in the specific part of the device, it will

affect the total setup.

III. PROPOSED SYSTEM

The complexity of the existing Braille system for the visually

impaired people is that it requires, the text to be translated to

Braille literature. Translating a book or a document into Braille literature is a complex, time consuming and more

expensive process. Day-to-day information cannot be translated into Braille literature. To ease this process of reading for the visually impaired people, this prototype has

been proposed. Benefits of the visually impaired are not

necessary to learn any new language. The visually impaired

can easily understand because different languages
converted

into native languages and hearable audio output. In the
Proposed system, all external devices are avoided. In
the

Existing System, only can convert specific languages
like Fig. 2: Block Diagram of Dynamic Web Content
Reader

Tamil and English. But in the proposed system,
establish The fig.2 is the block diagram of the Web
Reader

multiple conversion of languages like English, Arab,
Application. This diagram shows the flow of the Web
Reader

Japanese, French etc. In the Proposed System,
convert the Application. The user can give the prefixed
URL input to the

dynamic web content using web mining and Natural Web Reader Application. And the python libraries like Language Processing concepts. It is easy to use and portable. requests, bs4, textblob are processed the URL and retrieve the

All kind of people can use this application. And content can specified web content from the given particular URL. And the

easily understand by the visually impaired persons also other retrieved web content is converted into the audio(.mp3)

language peoples are able to understand the different format in native language Tamil.

languages in our native languages.

IV. DESIGN & IMPLEMENTATION

Our designed Project is called Web Reader, a simple

application with the text to speech functionality. The system

was developed using web mining and NLP concepts using python.

The application is divided into three modules - the main application module which includes the basic GUI components which handles the basic operations of the application such as input of parameters for conversion either via file or direct keyboard input.

The second module, Requesting the targeted website using urllib to fetch by using GET method and read the content as object (page as Object). Separate content from

DOM object by using bs4 and find the targeted object.

The third module, Converting DOM object as a Fig. 3:
sample output for Web Reader Application

string. Parse string to the textblob. Finally Set output

The python library requests, bs4, textblob, gtts,
language & store in array (language conversion) and
loop the pydub, lxml, os are used in our project. The
detailed explanation

string converted to audio. of each library is followed by,

Web Reader converts Dynamic web content to

A. Requests

Audio output in native Language in the application. It

Requests will allow you to send HTTP/1.1 requests
using

provides multiple language conversion into native
language.

Python. With it, you can add content like headers, form
data,

Web Reader loads the content from the website and the

multipart files, and parameters via simple Python libraries. It

Language Conversion procedure starts automatically. also allows you to access the response data of Python in the

Web Reader contains an exceptional function that same way.

gives the user the choice of saving its already converted text

to any part of the local machine in an audio format; this allows B. Bs4

the user to copy the audio format to any of his/her audio

BeautifulSoup 4. BeautifulSoup is a Python library for

devices, so that they can hence forth treat it as an audio book.

pulling data out of HTML and XML files. It works with your

favorite parser to provide idiomatic ways of navigating,

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searching, and modifying the parse tree. It commonly saves

programmers hours or days of work

C. Textblob

TextBlob is a Python (2 and 3) library for processing textual

data. It provides a simple API for diving into common natural

language processing (NLP) tasks such as

part-of-speech

tagging, noun phrase extraction, sentiment analysis,

classification, translation, and more.

D. gtts

gTTS (Google Text-to-Speech), a Python library and

CLI

tool to interface with Google Translates text-to-speech

API.

Writes spoken mp3 data to a file, a file-like object

(bytestring) for further audio manipulation, or stdout. It

Fig. 5: sample output for bs4 library

features flexible pre-processing and tokenizing, as well

as

The bs4 library is used to Converting the DOM object

as a

automatic retrieval of supported languages.

string. The converted string is given to the `textblob` library it's

E. `pydub` convert the text input to audio(.mp3) format output.

Manipulate audio with an simple and easy high level interface.

F. `lxml`

The `lxml` XML toolkit is a Pythonic binding for the C libraries `libxml2` and `libxslt`. It is unique in that it combines

the speed and XML feature completeness of these libraries

with the simplicity of a native Python API, mostly compatible but superior to the well-known `ElementTree` API.

G. OS

The OS module in Python provides a way of using operating

system dependent functionality. The functions that the OS

module provides allows you to interface with the underlying

operating system that Python is running on – be that

Fig. 6: sample output of the Web Reader Application

Windows, Mac or Linux.

Initially four languages (India-English, USA-

English, Arab, Japan) are converted into native languages and

in future many language's can be converted into native language.

V. CONCLUSION

Web Reader Application is a flexible robust dynamic growing aspect of modern computer era and it is increasingly

playing a more significance role in the way we interact with

the system and interfaces which is based on platform independent concept. We have identified the various operations and processes involved in text to speech synthesis.

We have also developed a very simple and attractive graphical user interface which allows the user to choose the

language in the application. Our system interfaces with a text

Fig. 4: sample output for requests library to speech engine developed for Arabic, Japanese, english. In

The Requests library used to Requesting the targeted future, we plan to make efforts to create engines for website using urllib to fetch by using GET method and read conversion of one language to other make text to speech

the content as object (page as Object). Separate content from technology more accessible to a wider range. Accuracy of the

DOM object by using bs4 and find the targeted object. software is excellent in the context of its ability to work in

real-life environment. We also have plans to make it a web

based real-time synthesis system, so that its uses can get more expanded.

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